

# Project Update

## Numerical Simulation of Manufacturing Process

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- ① Progress
- ② Currently working on
- ③ Future work

① Progress

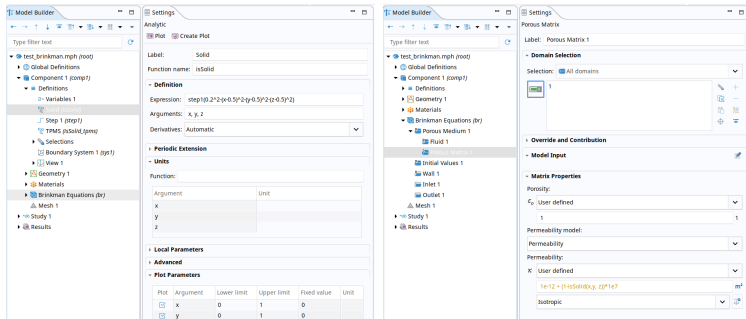
② Currently working on

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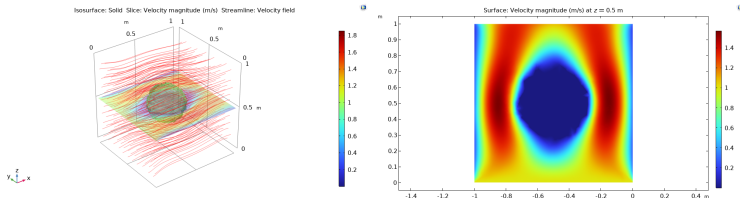
# Sphere example

Before considering a TPMS structure, we have to be familiar with Comsol and Brinkman penalization. For Axel, this means starting with a more simple example.

- Define an analytic function for the obstacle
- Define the porous medium properties for the region



# Sphere example results



Conclusion: Brinkman frameworks works for very simple geometries

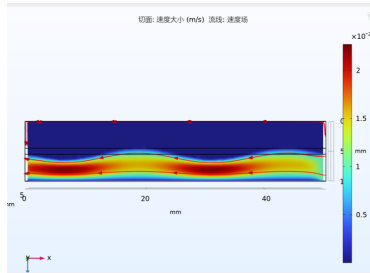
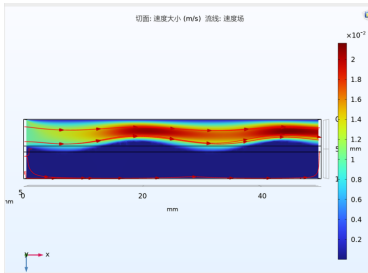
# TPMS analysis

The gyroid TPMS structure can be implemented in the same way as the sphere, i.e. by an analytic expression.

① Progress

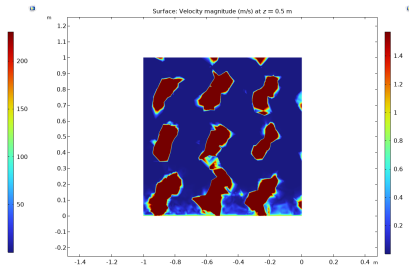
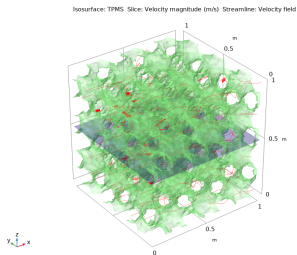
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① Progress

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③ Future work

- Incorporate heat transfer into the model
- Successfully implement a two-channel model